the Nansen LEGACY

Annual Meeting

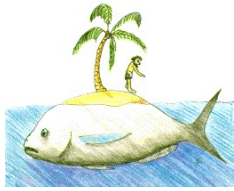
01 – 03. November 2022

Evaluating a climate-driven spatial food web model for the Barents Sea by comparison with field surveyed spatial distributions for functional groups

Marcela C. Nascimento, Bérengère Husson, Lilia Guillet, Torstein Pedersen

THE ARCTIC UNIVERSITY OF NORWAY • UIT •

Main questions

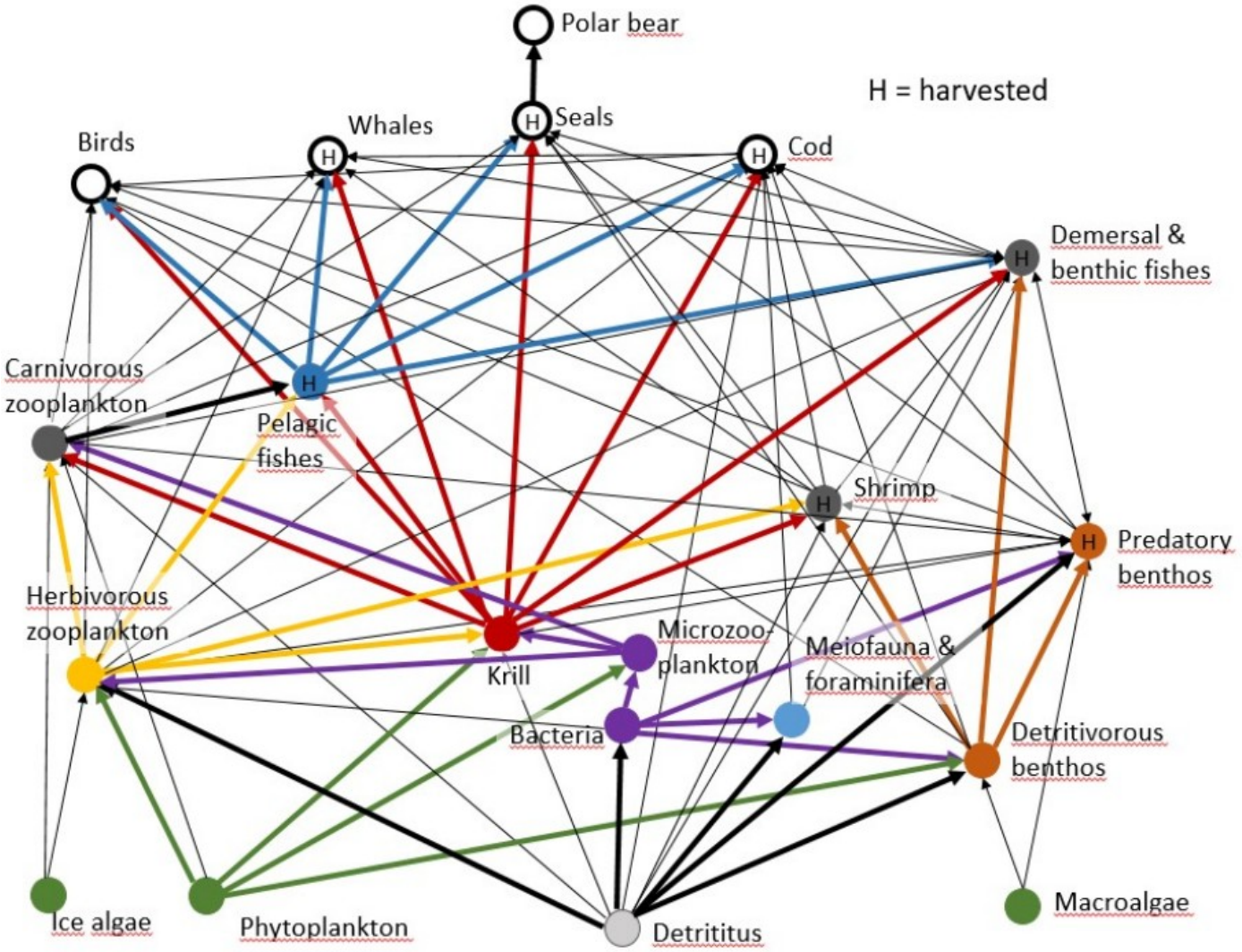


Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

frontiers in Marine Science

ORIGINAL RESEARCH
published: 17 September 2021
doi: 10.3389/fmars.2021.732637



Trophic flow

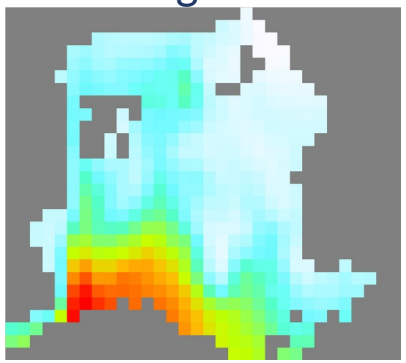
Overexploitation, Recovery, and Warming of the Barents Sea Ecosystem During 1950–2013

Torstein Pedersen[1], Nina Mikkelsen[1,2], Ulf Lindstrøm[1,2], Paul E. Renaud[3,4], Marcela C. Nascimento[1], Marie-Anne Blanchet[5†], Ingrid H. Ellingsen[6], Lis L. Jørgensen[2] and Hugues Blanchet[3,7]*

High resolution mass balanced model with 108 Functional groups (FG)

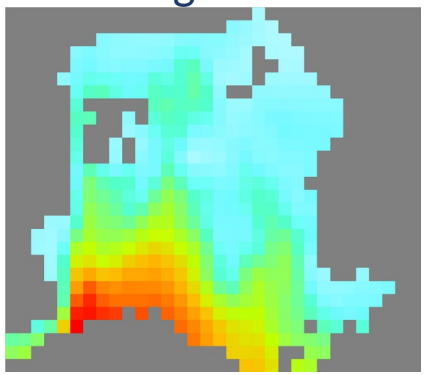
2004 Bottom T°C

Average 0.62 C



2013 Bottom T°C

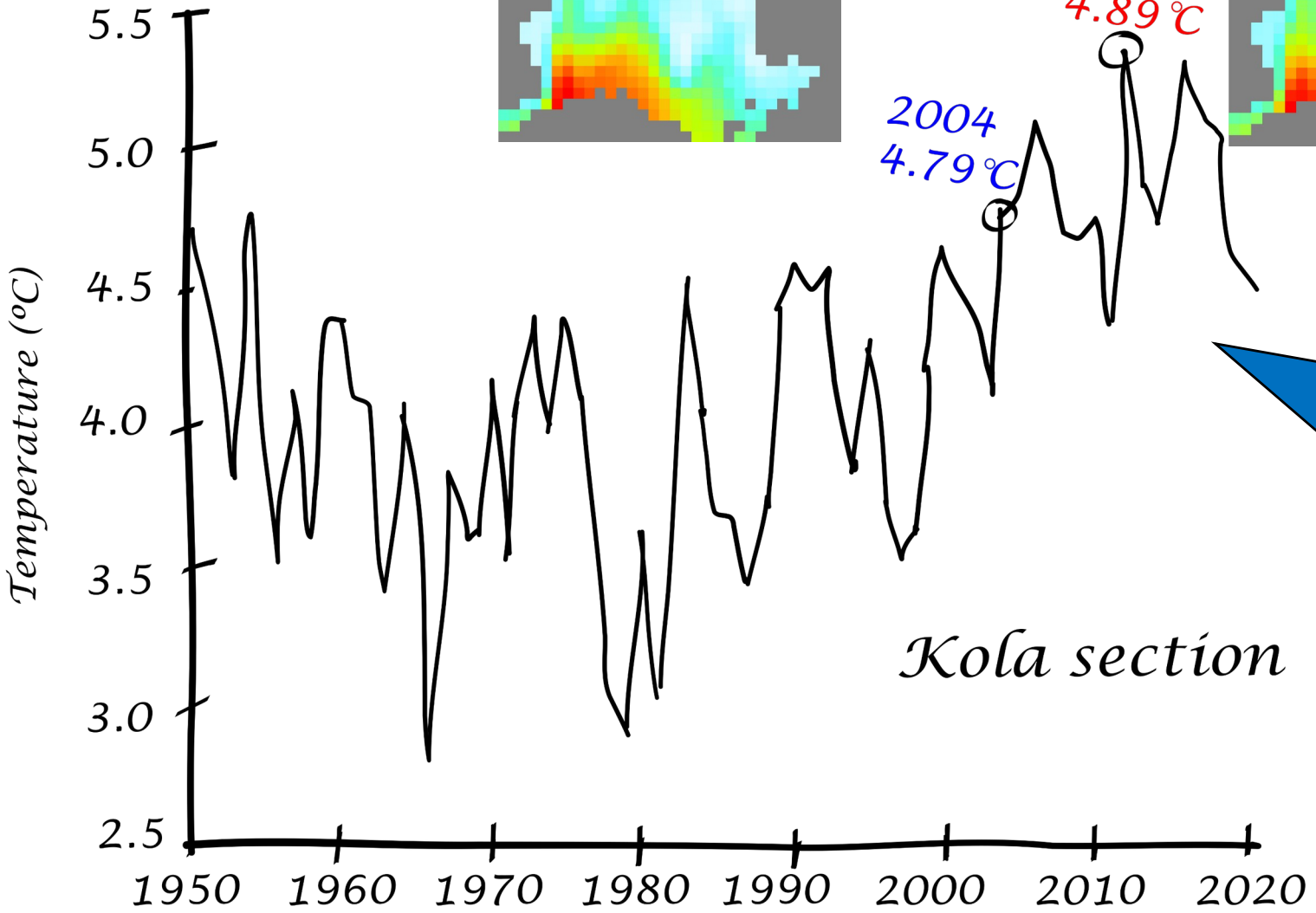
Average 0.89 C

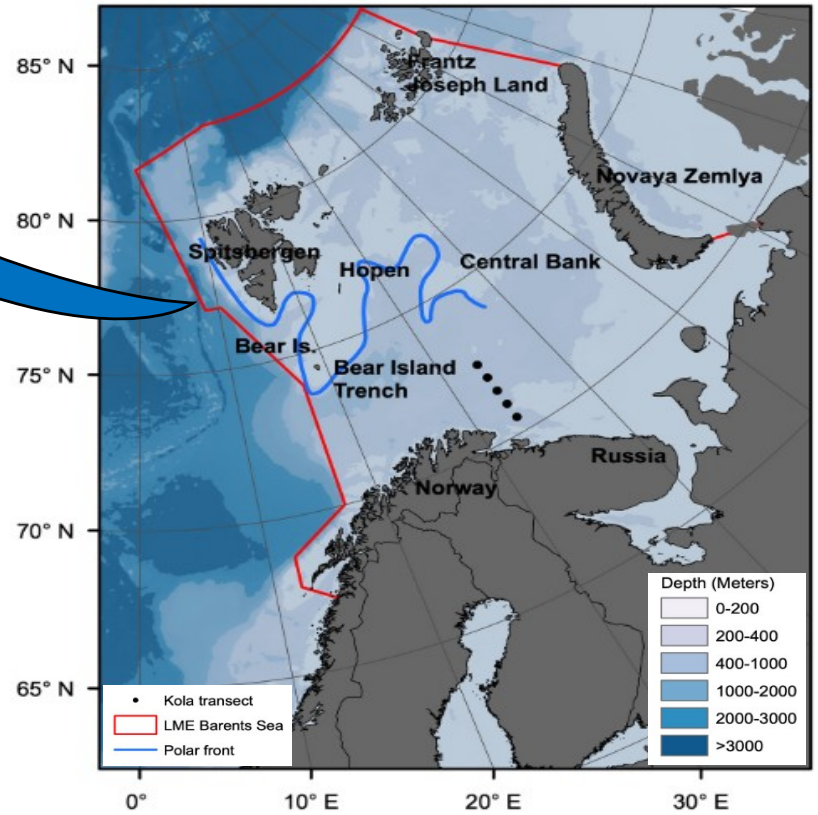


SVIM archives
Regional model
historical data

Bottom temperature
Surface temperature
Ice coverage

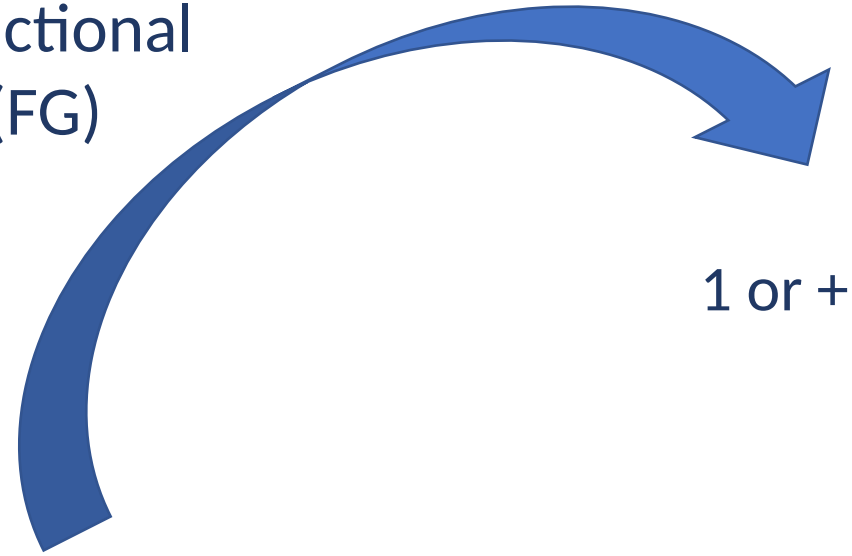
Primary production





Habitat capacities

108 Functional groups (FG)



74 FG
1 or + environmental response

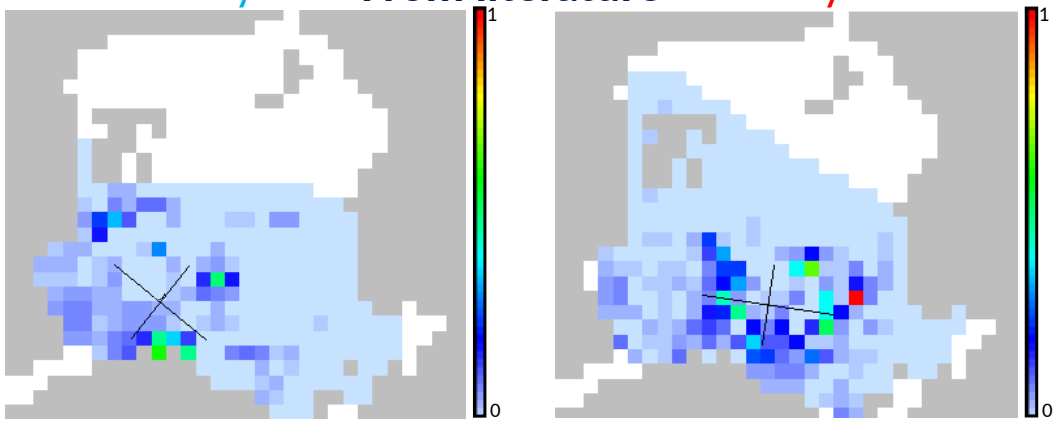
QGAM model – ecosystem survey + literature

Functional groups with high biomass and good distribution information
35 FGs

Biomass distribution observed vs modelled

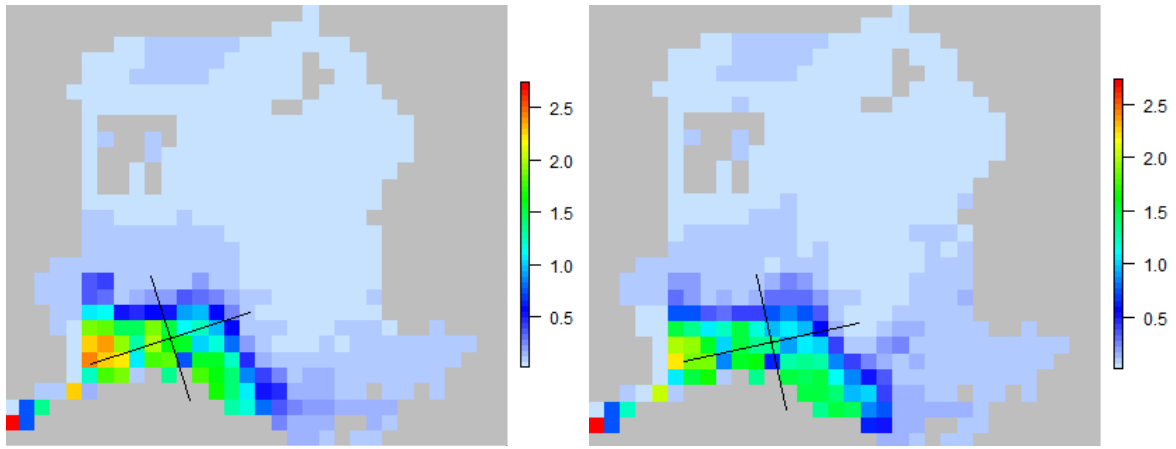
Observed

Cold year **From literature** Warm year

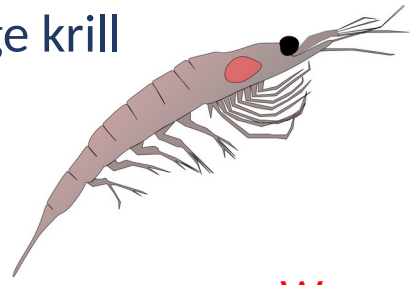


Mikhina et al. (2019), Eriksen and Dalpadado (2011) and Orlova et al. (2010)

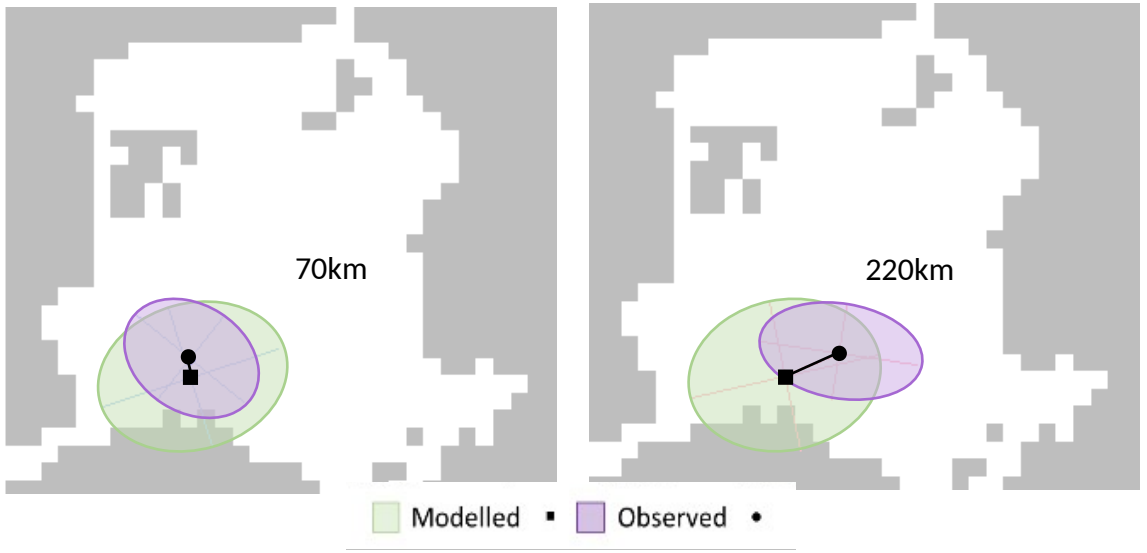
Modelled



Large krill



Cold year Warm year

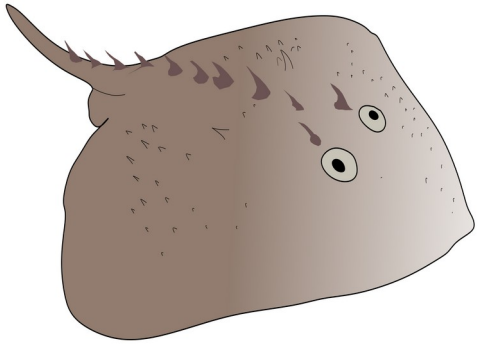


Average CG of all FG

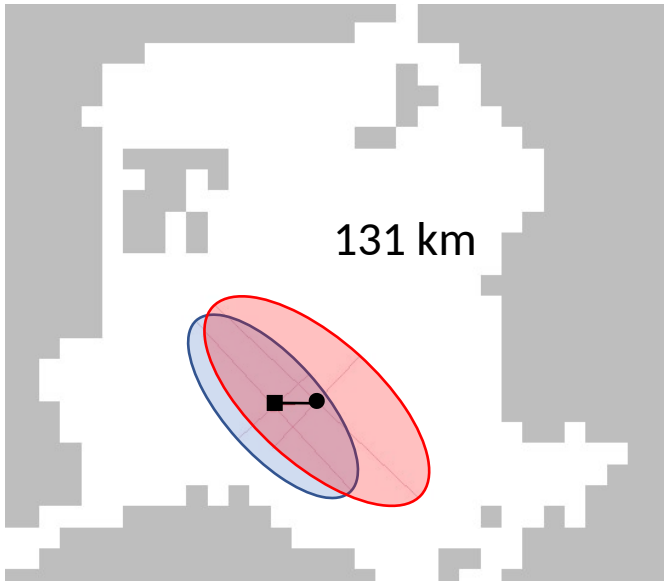
Cold year **19** km
Warm year **26** km

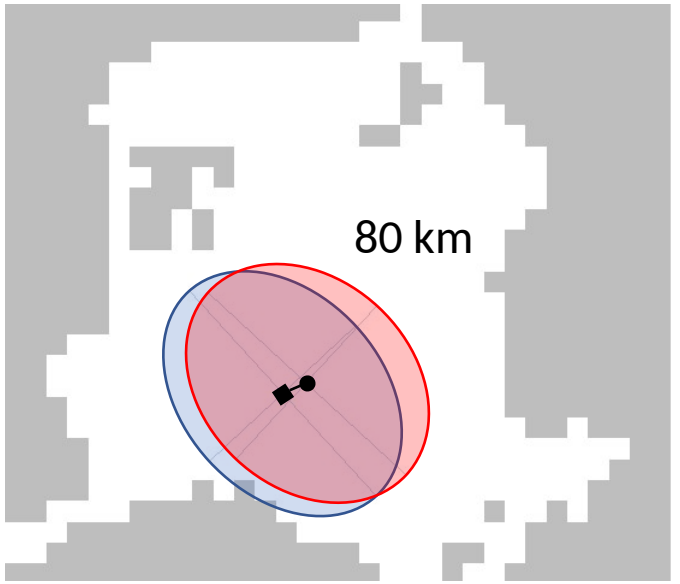
Shift in distributions from cold to warm year

35 FGs



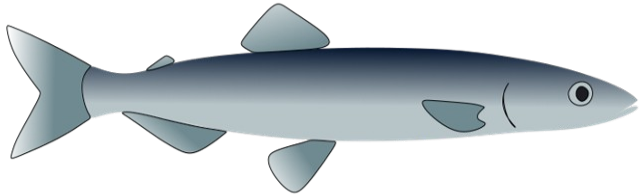
Thorny skate



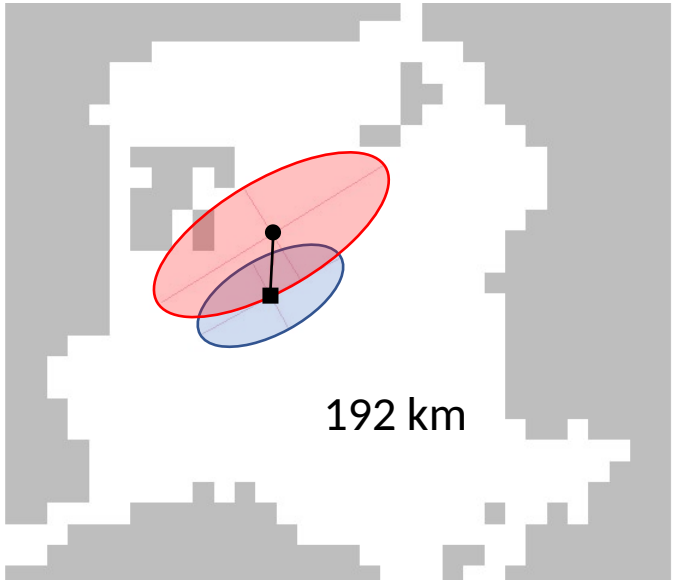


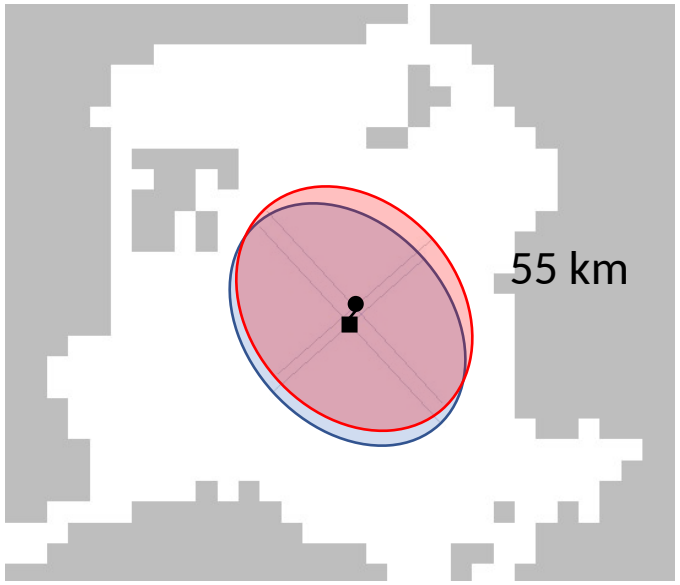
Observed

Modelled



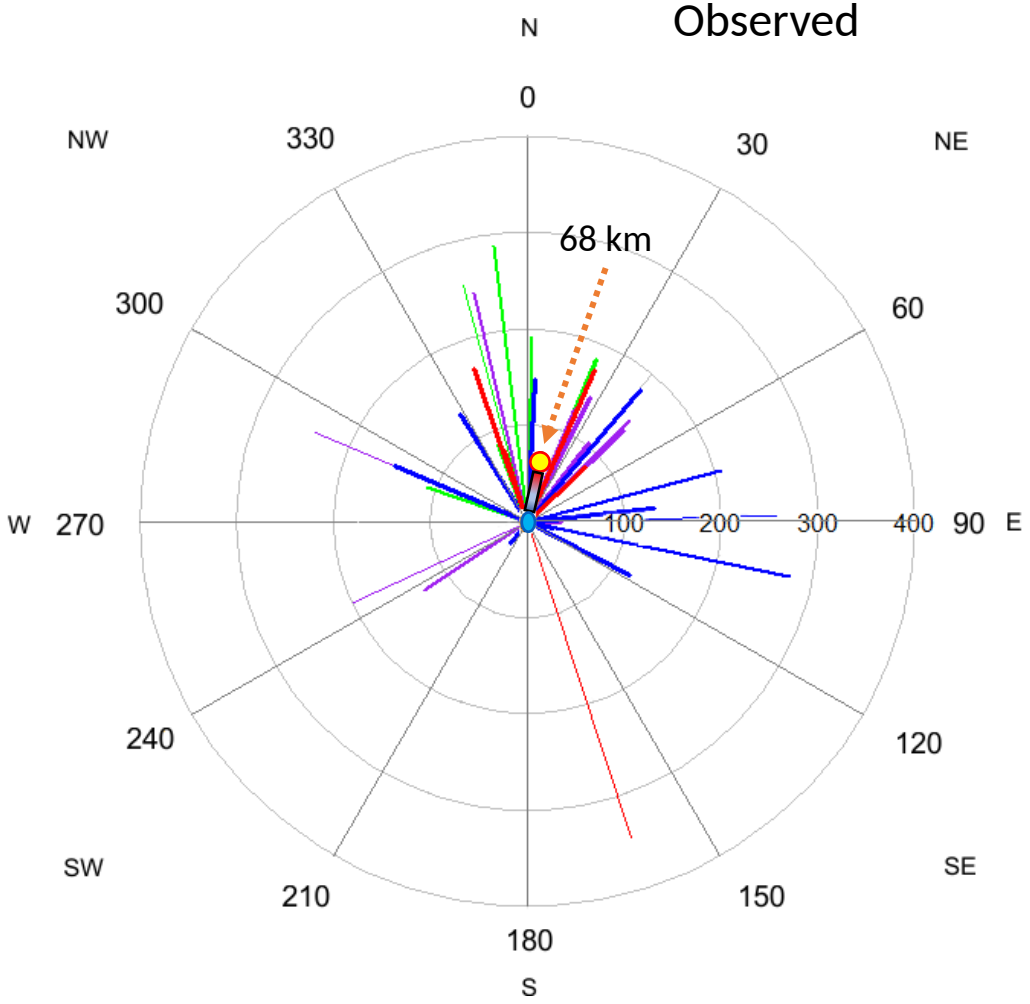
Capelin (3+)



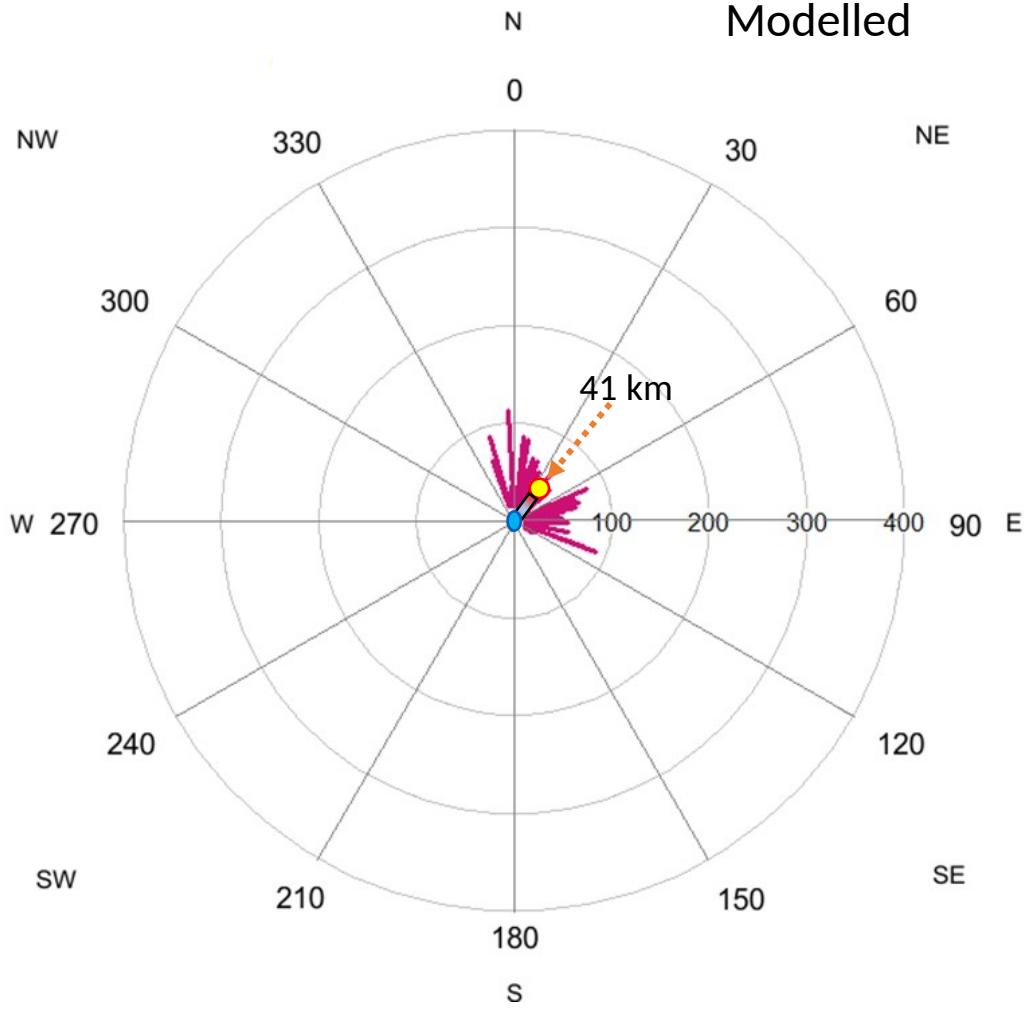


Shift in distributions from cold to warm year for 35 FG

Observed



Modelled



Average directed shift in CG position

Summarizing

Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

The model can represent well the FGs distributions.

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

The model can predict well the direction of shift, mainly north-eastern

The CG average positions, the displacement direction and the distance predicted by the model are very similar to the survey based observations.

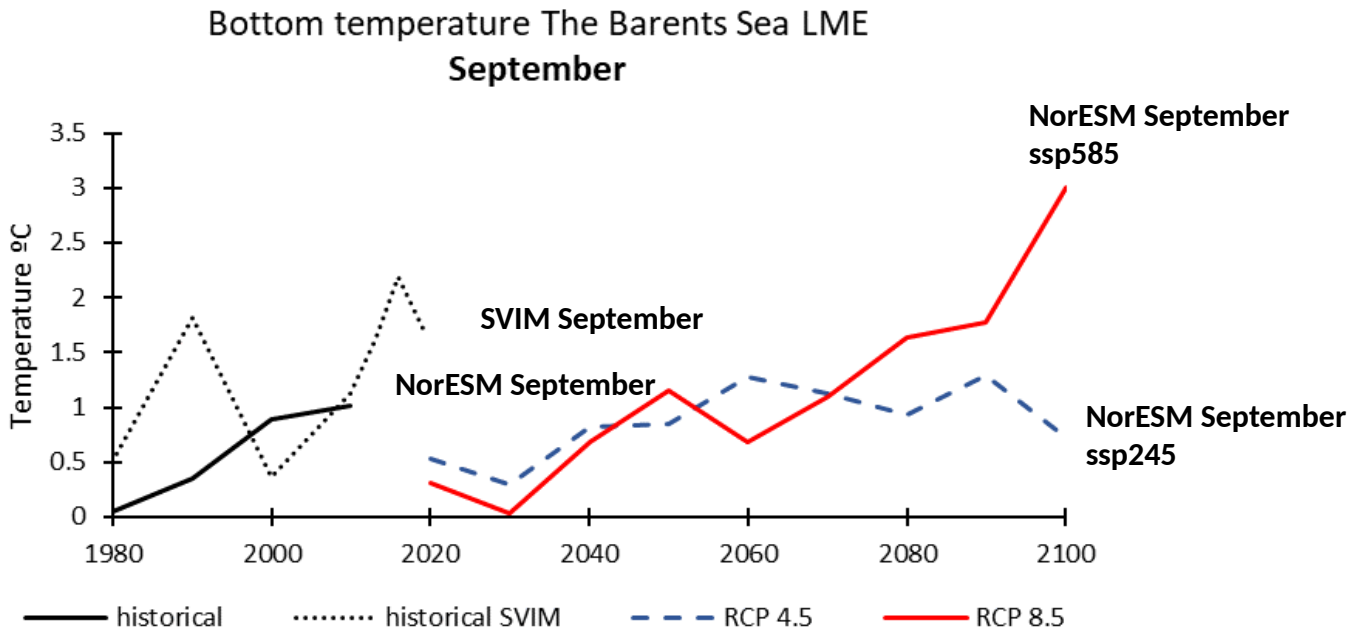
Manuscript in review

What's next?

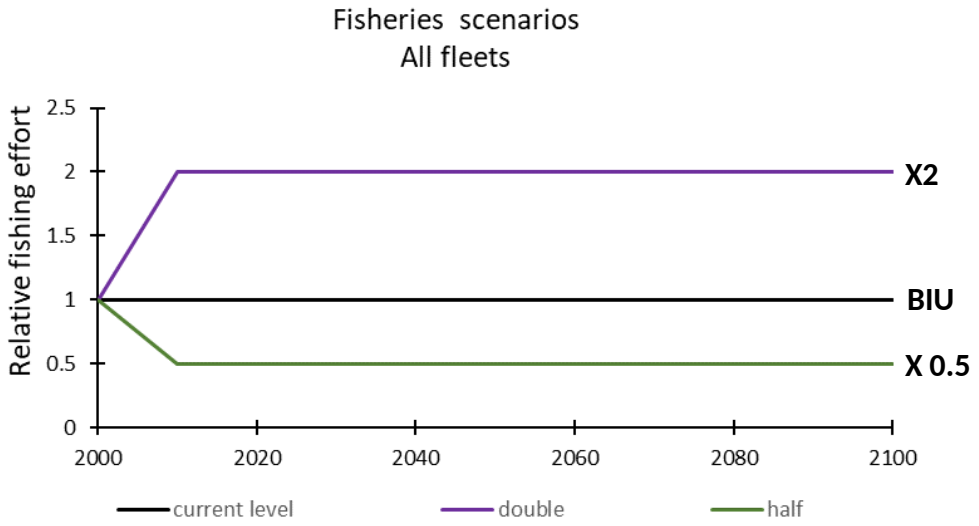
NorESM-MM

Goals are to investigate future climate changes effects on spatial distributions

Environmental changes
Based on RCP 4.5 and RCP 8.5



Fisheries scenarios



Exploratory run

Ice coverage, bottom and surface temperature scenarios

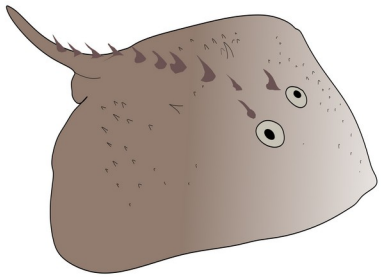
Annual average 2050

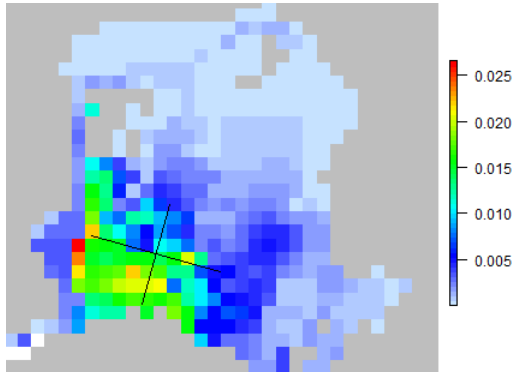
Current level NorRESM cool

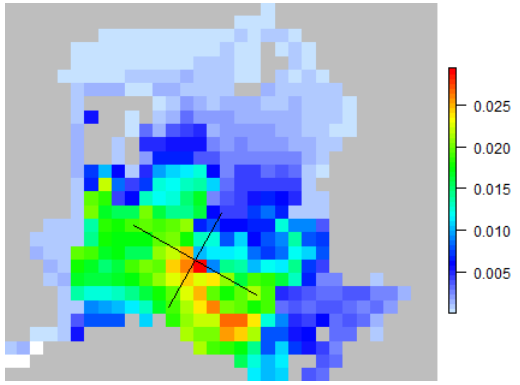
RCP 4.5 warm

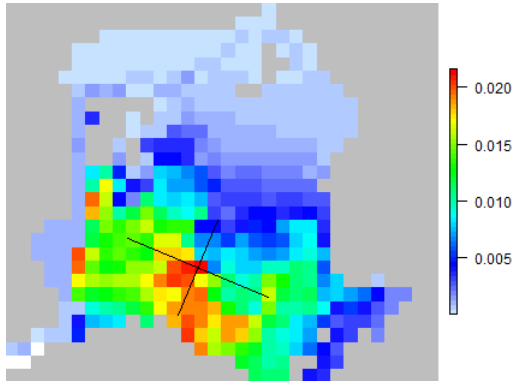
RCP8.5 warmest

Thorny skate

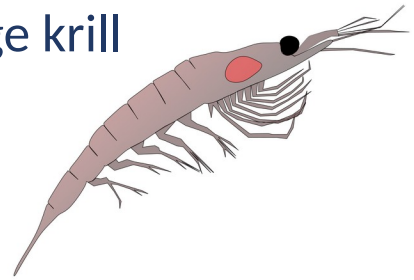


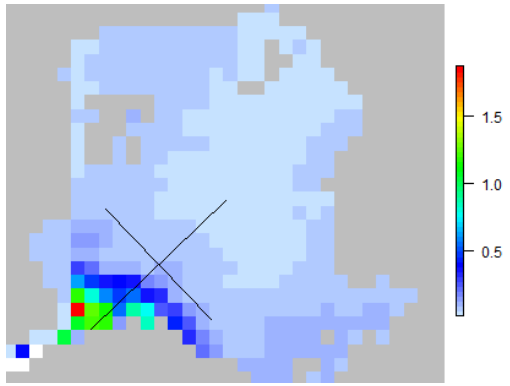


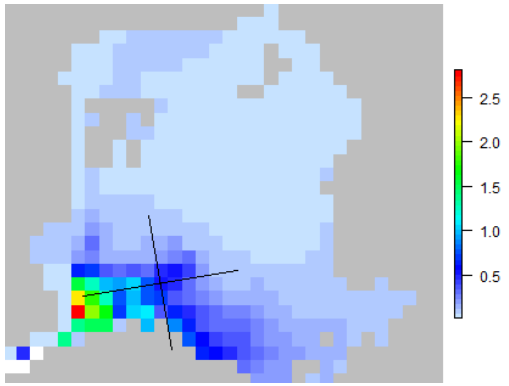


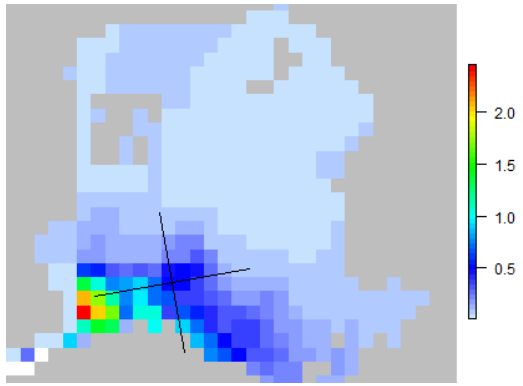


Large krill









Relative biomass